management schedule and minor inputs

The management schedules are in `management.sch`. Four small inputs referred to from other pages follow it: `puddle.ops`, `transplant.ops`, `co2_yr.dat` and `object.prt`.

management.sch

`management.sch` contains the management schedules. Each schedule has a list of operations on fixed dates or heat-unit fractions, a list of automatic operations (decision tables in `lum.dtl`), or both. An HRU uses the schedule named in the `mgt` column of its land use in `landuse.lum`. The file name is the second entry on the `lum` line of `file.cio` (default `management.sch`).

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Then, for each schedule:

1. The schedule line: `name`, `numb_ops`, `numb_auto`.
2. `numb_auto` lines, each naming one decision table in `lum.dtl`.
3. `numb_ops` operation lines.

Column	Description
`name`	Schedule name, referenced by `mgt` in `landuse.lum`.
`numb_ops`	Number of scheduled operation lines.
`numb_auto`	Number of automatic operation lines.

Automatic operations

Each automatic operation line names a decision table in `lum.dtl`. Every day, SWAT+ checks each listed table for every HRU that uses the schedule. A name that is not in `lum.dtl` is not reported.

Three generic tables can be shared by several schedules with different crops. For these, the crop name follows the table name on the same line, and the `plant`, `harvest` and `harvest_kill` actions of the table use it instead of their `option` column:

Table name	Line content
`pl_hv_summer1`	table name and one crop
`pl_hv_winter1`	table name and one crop
`pl_hv_summer2`	table name and two crops; the crop used is the one for the current rotation year

Operation lines

#	Column	Units	Description
1	`op_typ`	-	Operation code (see below).
2	`mon`	-	Month of the operation, or 0.
3	`day`	-	Day of the month, or 0.
4	`hu_sch`	fraction	Heat-unit fraction that triggers the operation, or 0.

#	Column	Units	Description
5	op_data1	-	First name (see below).
6	op_data2	-	Second name (see below).
7	op_data3	varies	Number: amount, days or override (see below).

op_typ	Operation	op_data1	op_data2	op_data3
plnt	plant	plant name in the HRU's plant community	name in transplant.ops	> 0 to start from the transplant entry
harv	harvest	plant name, or all	name in harv.ops	residue harvest efficiency (0 uses harv_eff)
hvkl	harvest and kill	plant name, or all	name in harv.ops	residue harvest efficiency
kill	kill	plant name, or all	-	-
till	tillage	name in tillage.til	-	-
fert	fertilizer	name in fertilizer.frt	name in chem_app.ops	amount, kg/ha
manu	manure	name in manure_db.frt	name in chem_app.ops	amount, kg/ha
pest	pesticide	pesticide name listed in constituents.cs	name in chem_app.ops	amount, kg/ha (multiplied by app_eff)
graz	grazing	name in graze.ops	-	number of grazing days
burn	burning	name in fire.ops	-	-
swep	street sweeping	name in sweep.ops	-	-
irrm	irrigation	name in irr.ops	-	-
irpm	irrigation of a rice paddy	name in irr.ops	-	-
irrp	start, adjust or stop paddy irrigation	name in irr.ops	water source: cha, res, aqu or unlim	target ponding depth, mm (see below)
pudl	puddling	name in tillage.til	name in puddle.ops	-
weir	set paddy weir height	weir name in weir.res	-	weir height, mm
dwm	drainage water management	-	-	tile drain depth, mm
cnup	change curve number	absval, abschg, pctchg or relchg	-	value
mons	monsoon season	-	-	0 or 1
skip	end of a rotation year	-	-	-

Notes on individual operations:

- plnt, harv, hvkl and kill act on the plant in the HRU's plant community (plant.ini) whose name equals op_data1. A plant that is already growing is not planted again; the event is logged as PLANT_ALREADY_GROWING.
- fert on an HRU with standing water puts the fertilizer into the water.
- pest looks the name up in the pesticide list of constituents.cs when the operation runs, so the pesticide must be simulated as a constituent.
- irrm and irpm apply the depth in irr.ops (see irr.ops).
- irrp turns on automatic irrigation of a paddy: water is added whenever the ponding depth falls below 90% of the target. A positive op_data3 is the target depth; a negative value uses amt_mm of the irr.ops entry; zero turns paddy irrigation off. The irr.ops entry also gives the efficiency, runoff fraction and salt and nitrate concentrations of the water.
- cnup changes the curve number for moisture condition II: set it to the value (absval), add the value (abschg), change it by the value in percent (pctchg), or multiply it by 1 + value (relchg). The result is kept between 35 and 95.
- mons applies to plants with a moisture_gro trigger in plants.plt. With op_data3 1 it marks the start of the monsoon season. With 0 it restarts the phenology of such plants that are not growing.
- Codes not in this table (for example pcom or prtp) are read but do nothing.

Timing

Operations run in the order of the lines, one rotation year after another. A skip line ends a rotation year. After the last line the schedule starts again at the first line, so a schedule with n skip lines is an n-year rotation.

An operation runs on the day its `mon` and `day` match the calendar date, or, when `hu_sch` is greater than 0, on the first day the heat-unit fraction exceeds `hu_sch`. The heat-unit fraction is that of the plant named in `op_data1` while it grows, and the base-zero fraction of the year otherwise. Give each operation either a date or a heat-unit fraction and set the other to 0. Within a rotation year, lines must be in chronological order: SWAT+ waits for the date of the current line before it moves on.

When the schedule reaches a `skip` line it waits there until the end of the year: 31 December for HRUs in the northern hemisphere, day 181 for HRUs in the southern hemisphere. At the start of the run, SWAT+ begins at the rotation year given by the plant community in `plant.ini` and skips the operations dated before the first simulated day.

The operations performed are written to `mgt_out.txt` when `mgtout` is `y` in `print.prt`.

Example

`refdata/Ames_sub1/management.sch` in the SWAT+ repository, first lines:

```
management.sch file AMES
                          NAME NUMB_OPS NUMB_AUTO OP_TYP        MON        DAY HU_SCH   OP_DATA1    OP_DAT
mgt_01                             305         1
                                                irr_year_irr
                                                    till          4          1    0.0    chisplow        n
                                                    plnt          4         28    0.0        corn        n
                                                    irrm          5         15    0.0        sprinkler_ilm
                                                    fert          6          1    0.0        null      ele
                                                    hvkl         10          7    0.0        corn       gr
                                                    skip          0          0      0        null        n
```

Schedule `mgt_01` has 305 operation lines and one automatic operation, the `irr_year_irr` table of `lum.dtl` (see the `*.dtl` example). In the first rotation year it tills with `chisplow` on 1 April, plants `corn` on 28 April, irrigates once on 15 May, and harvests and kills the corn with the `grain` harvest operation on 7 October.

For a `fert` line, 62.0.1 reads the fertilizer name from `op_data1` and the application method from `op_data2`. The `fert` line above has `null` in `op_data1` and the fertilizer name `elem_n` in `op_data2`. No fertilizer is named `null`, so the line matches no entry in `fertilizer.frt` and applies nothing; no warning is written. Every `fert` line in this file has the same layout, so as written the Ames schedules apply no fertilizer in 62.0.1. A line that applies 181.4 kg/ha of `elem_n` in 62.0.1 reads `fert 6 1 0.0 elem_n broadcast 181.4`, with `broadcast` an entry in `chem_app.ops`.

Two schedules from `refdata/Osu_1hru/management.sch`, one using a scheduled `manu` operation together with decision tables and one using only decision tables:

```
rice140_rot                        1           7
     plow
     weir60r
     weir0
     pad_irr_r
     puddle
     pl_hv_rice
     fert_paddy
manu       5          1       0     beef_salt    broadcast   12.9927
agrl_rot                           0          2
    pl_hv_agro
    fert_stress
```

As shipped, the `manu` line finds `beef_salt` in `manure_db.frt` and `broadcast` in `chem_app.ops`, so 12.99 kg/ha of manure is applied on 1 May. The schedule has one operation line and no `skip`, so the line runs every year.

Related files

- `landuse.lum`: selects a schedule by name in its `mgt` column.
- `*.ops`, `tillage.til`, `fertilizer.frt`: databases the operation lines name.

- *.dtl: the decision tables listed as automatic operations.
- plant.ini: plant community and initial rotation year.

puddle.ops

Puddling parameters for rice paddies. A puddling operation (`pudl` in `management.sch`, or the `puddle` action in lum.dtl) mixes the soil with a tillage operation, lowers the hydraulic conductivity of the paddy's upper soil, and sets the sediment concentration of the ponded water. The file name is fixed.

Free format. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one entry, read by position.

#	Column	Units	Description
1	`name`	-	Puddling operation name.
2	`hydcon_mm/h`	mm/h	Hydraulic conductivity of the upper soil after puddling. 0 leaves it unchanged.
3	`sed_ppm`	ppm	Sediment concentration of the ponded water after puddling.
4	`orgn_ppm`	ppm	Organic N concentration after puddling. Read but not used.
5	`sedp_ppm`	ppm	Organic P concentration after puddling. Read but not used.
6	`no3_ppm`	ppm	NO3-N concentration after puddling. Read but not used.
7	`solp_ppm`	ppm	Soluble P concentration after puddling. Read but not used.
8	`nh3_ppm`	ppm	NH3 concentration after puddling. Read but not used.
9	`no2_ppm`	ppm	NO2 concentration after puddling. Read but not used.

The tillage operation is named in `op_data1` of the `pudl` line (or the `name` column of the `puddle` action). If the paddy has more than 1 mm of water, the mixing is done in the ponded water; otherwise it is ordinary tillage.

`refdata/Osu_1hru/puddle.ops` as shipped:

name	hydcon_mm/h	sed_ppm	orgn_ppm	sedp_ppm	no3_ppm	solp_ppm	nh
high_eff	0.00000	10000	0.00000	800.00000	10.00000	20.00000	1
med_eff	0.00000	10000	0.00000	500.00000	10.00000	20.00000	1
low_eff	0.0	10000	0.00000	100.00000	10.00000	20.00000	1.000

The file has no title line. SWAT+ skips the column header as the title and the `high_eff` line as the header, so 62.0.1 reads only `med_eff` and `low_eff`. An operation that names `high_eff` finds no entry and uses an entry with all values zero: conductivity unchanged and no sediment added. The Osu dataset's `puddle` decision table names `med_eff`, so its puddling runs with a sediment concentration of 10000 ppm and unchanged conductivity (`hydcon_mm/h` is 0).

transplant.ops

Transplant database. A planting can start a plant from a transplanted state instead of from seed: a `plnt` line in `management.sch` with a `transplant.ops` name in `op_data2` and `op_data3` greater than 0, or a `plant` action in lum.dtl with the name in its `fp` column. The file name is fixed. Up to revision 61 it was `transplant.plt`; 62.0.1 reads only `transplant.ops`, so a project with a `transplant.plt` file needs it renamed.

Free format. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one entry, read by position.

#	Column	Units	Description
1	`name`	-	Transplant name.
2	`lai_ini`	m^2/m^2	Leaf area index at transplanting. Read but not used.

#	Column	Units	Description
3	bm_ini	kg/ha	Plant biomass at transplanting.
4	phu_acc_ini	fraction	Fraction of potential heat units already accumulated.
5	fr_yrmat	fraction	Fraction of the years to maturity already reached (perennials).
6	pop	plants/m²	Plant population. Sets the maximum leaf area index through the population curve in plants.plt. 0 uses the maximum LAI in plants.plt.

refdata/Osu_1hru contains the file under its old name:

transplant.plt

NAME	LAI_INI	BM_INI	PHU_ACC_INI	FR_YRMAT	POP
tr_rice140	0.1	0.2	0.2	1	100

62.0.1 does not open transplant.plt, so in this dataset the transplant database is empty. The plant actions in its lum.dtl that name tr_rice140 find no entry, no message is written, and the rice is planted from seed with the initial values of an ordinary planting.

co2_yr.dat and co2.out

Atmospheric CO2 concentration by year. co2_yr.dat is optional; its name is fixed. SWAT+ builds one CO2 value per simulation year and writes the series to co2.out.

co2_yr.dat layout:

Line	Content
1	title (skipped)
2	number of year records n
3	header (skipped)
4 to $n + 3$	calendar year and CO2 concentration (ppm), one year per line, consecutive years in increasing order

How the series is built:

- If co2_yr.dat is missing or empty, every year uses co2 from parameters.bsn (default 400 ppm).
- If the last record's year is at or before the first simulation year, every year uses the last record's value.
- Otherwise the records are aligned to the simulation by the first record's year. Years before the first record use the first value, and years after the last record repeat the last value.

The alignment counts records from the first one, so the years must be consecutive.

co2.out has the header YR CO2(ppm) and one line per simulation year with the calendar year and the CO2 value used.

object.prt

Hydrograph and state output for individual objects. Each line asks for one variable of one object, written daily to a separate file. The file name is the third entry on the simulation line of file.cio (default object.prt; null for none).

Free format. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one request.

#	Column	Description
1	id	Sequence number, 1, 2, 3, … in order.
2	obj_typ	Object type: hru, hlt (HRU-lte), ru (routing unit), res, cha (channel), sdc (SWAT-DEG channel), exc (export coefficient), dr (delivery ratio) or out (outlet).

#	Column	Description
3	obj_typ_no	Object number within its type. For hru with sol_water or solnut_ly, 0 writes every HRU.
4	hyd_typ	Variable (see below).
5	filename	Output file name, up to 26 characters.

hyd_typ	Output
tot	total outflow hydrograph
rhg	recharge (percolation) hydrograph
sur	surface runoff hydrograph
lat	lateral flow hydrograph
til	tile flow hydrograph
sol_water	soil water of each layer
solnut_ly	soil N and P pools of each layer (no header line)
solnut_pr	soil N and P pools of the profile (no header line)
plant	plant status
cha_fp	channel and floodplain water balance

Each requested file gets a header line (except solnut_ly and solnut_pr) and one line per day with day of year, month, day, year, object type and name, followed by the values.

Sources

- management.sch readers: mgt_read_mgtops.f90 (schedule and automatic operation lines), read_mgtops.f90 (operation lines and name lookups)
- Schedule types: mgt_operations_module.f90
- Running scheduled operations: mgt_operatn.f90, mgt_sched.f90, called from hru_control.f90
- Starting operation and year-end skip handling: plant_init.f90, time_control.f90
- Automatic operation lookup: proc_cond.f90; generic crop tables: actions.f90
- Curve number change: chg_par.f90
- puddle.ops: mgt_read_puddle.f90
- transplant.ops: plant_transplant_read.f90, mgt_transplant.f90, type plant_transplant_db in plant_data_module.f90
- CO2: co2_read.f90, bsn_prm%co2 in basin_module.f90
- object.prt: object_read_output.f90, obj_output.f90, type object_output in hydrograph_module.f90
- File name slots: in_lum%management_sch and in_sim%object_prt in input_file_module.f90

File	Column	Fortran variable
management.sch	numb_ops, numb_auto	sched%num_ops, sched%num_autos
management.sch	automatic operation line	sched%auto_name, sched%auto_crop
management.sch	op_typ, mon, day	mgt_ops%op, %mon, %day
management.sch	hu_sch	mgt_ops%husc
management.sch	op_data1, op_data2, op_data3	mgt_ops%op_char, %op_plant, %op3
puddle.ops	hydcon_mm/h	pudl_db%wet_hc
transplant.ops	lai_ini, bm_ini, phu_acc_ini	transpl%lai, %bioms, %phuacc
co2_yr.dat	year, CO2	co2_inc%co2_yr%iyr, %co2
object.prt	obj_typ, obj_typ_no, hyd_typ	ob_out%obtyp, %obtypno, %hydtyp